

8D Report: Bore Diameter Oversize on Valve Body AP-3312

Monterrey Plant, CNC lathe MTY-LTH-04, customer Brightline Hydraulics

Document ID	8D-2026-014
Revision	Final
Effective	Opened 14 August 2026, closed 18 September 2026
Owner	Quality Engineering, Monterrey Plant
Applies to	Part AP-3312 (Proportional valve body PV-12)

D0. Summary

On 14 August 2026 Brightline Hydraulics returned 48 proportional valve bodies (AP-3312) because the main bore was oversize. Between 3 August and 21 August, 12 further bore-diameter nonconformances (285 pieces) were recorded in process on the same part, all on CNC lathe **MTY-LTH-04** at the Monterrey Plant. The cost of the nonconforming parts, including the customer return, was \$12,844.80.

The root cause was a worn spindle bearing. The bearing's vibration rose for several weeks while the machine's PM-2 service was 12 days overdue, and the in-process gauge plan was too sparse to catch the drift. The spindle bearing failed on 24 August 2026 and the spindle cartridge was replaced. No further bore-diameter nonconformances were recorded after the repair.

This is a synthetic document written for a software demonstration. Atlas Precision Components, Brightline Hydraulics and every person named here are fictional.

D1. Team

Role	Name	Function
Champion	Elena Marsh	Plant Manager, Monterrey
Team leader	Rafael Ocampo	Quality Engineering Manager, Monterrey
Member	Dario Quintero	Maintenance Supervisor, Monterrey
Member	Ines Valdivia	CNC Programmer, Monterrey
Member	Tomas Reyes	Process Engineer, Monterrey
Customer contact	Helen Park	Supplier Quality Engineer, Brightline Hydraulics

D2. Problem description

Question	Answer
What	Main bore of valve body AP-3312 (proportional valve body PV-12) oversize; bore specification 22.000 mm, +0.021 / 0 mm.
Where	Monterrey Plant, CNC lathe MTY-LTH-04 (Cell D). Found at in-process gauging and by the customer.
When	Customer return detected 14 August 2026; in-plant records from 3 August to 21 August 2026.
Who	Monterrey operators and quality inspectors; Brightline Hydraulics incoming inspection.
How many	48 pieces returned by the customer; 285 pieces in 12 in-plant records.
How detected	Customer incoming inspection; in-process air gauge on one piece in 25.
Cost	\$12,844.80 of scrap and rework, including the customer return.

Table 1. Problem statement (5W2H).

The in-plant nonconformances were all recorded as defect type dimension_oob against part AP-3312 on lathe MTY-LTH-04, under work orders WO-00725 and WO-00784. Of the 285 pieces affected, 96 were scrapped and 189 were reworked.

D3. Interim containment actions

- 15 August: all AP-3312 stock at Monterrey (finished goods and work in progress) was quarantined and 100 percent checked with an air gauge.
- 15 August: every AP-3312 piece produced on MTY-LTH-04 is gauged at the machine until further notice, instead of one in every 25 pieces.
- 17 August: Brightline Hydraulics sorted the remaining pieces from the affected lot at its own site; no further nonconforming pieces were found.
- 19 August: the operator was told to stop the machine for any bore reading above 22.015 mm and call the maintenance supervisor.

D4. Root cause analysis

Evidence from the machine data

Daily vibration, temperature and alarm counts for MTY-LTH-04 show the bearing wearing out. In June and July 2026 the machine averaged 2.4 mm/s (the other CNC lathes' weekly median was 2.1 mm/s). From the week of 3 August vibration rose every week, to 5.4 mm/s in the week of 17 August, and the maximum temperature rose from 54 to 61 degrees C. The machine logged 19 alarms between 3 and 21 August. The PM-MAN-001 warning level for CNC lathes (3.5 mm/s) was first exceeded on three consecutive operating days on 10 August 2026 and the alarm level (4.5 mm/s) on 17 August 2026.

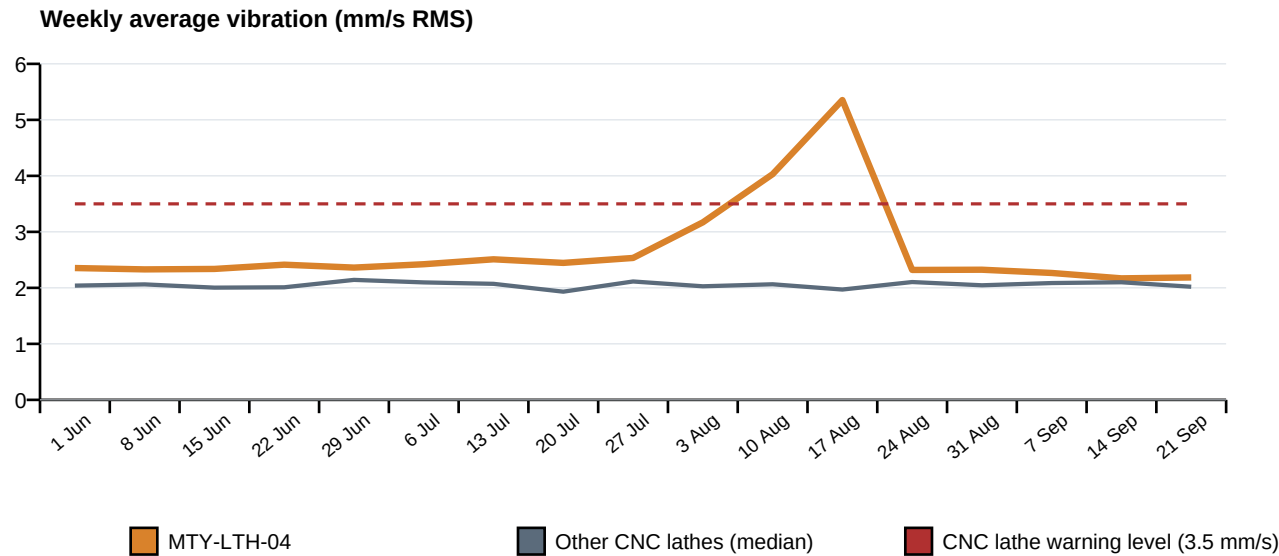


Figure 1. Weekly average vibration, MTY-LTH-04 against the median of the other CNC lathes (mm/s RMS).

Five whys

Why?	Answer
Why was the bore oversize?	The spindle ran out of true and the cutting position drifted as the bearing wore.
Why did the bearing wear out?	The bearing reached the end of its life while the lubrication and preload checks of PM-2 were not done on time.
Why was PM-2 late?	The last PM-2 was on 14 April 2026. The 120-day interval for CNC lathes (PM-MAN-001, Table 1) made it due on 12 August 2026. The planner deferred it twice to protect the Brightline delivery.
Why was the drift not caught earlier?	The vibration warning level was exceeded for 14 days before the failure but the alert went to a shared mailbox that nobody owned.
Why did the nonconforming parts leave the plant?	The gauge plan checked one piece in 25, which cannot detect a drift of a few microns an hour.

Root causes

Type	Root cause
Occurrence	Spindle bearing wear on MTY-LTH-04, failure code SPN-BRG.
Detection	Gauge frequency of one piece in 25 was insufficient for a drifting process.
Systemic	The CMMS allowed work orders to be released on a machine more than 14 days past its PM-2 interval, and vibration alerts had no owner.

D5. Permanent corrective actions

Action	Owner	Completed
Replace the spindle cartridge on MTY-LTH-04 (unplanned event, failure code SPN-BRG, 52.0 hours of downtime).	Dario Quintero	24 August 2026
Complete the overdue PM-2 on MTY-LTH-04.	Dario Quintero	9 September 2026
Route vibration and temperature warning alerts to the cell technician and the planner by name.	Ines Valdivia	28 August 2026
Gauge every fifth piece, and every piece for the first 50 after a tool change, on AP-3312.	Tomas Reyes	31 August 2026

D6. Verification of corrective actions

After the repair the machine's average vibration was 2.2 mm/s in September, in line with its baseline. No dimension_oob records were logged on MTY-LTH-04 between 25 August and 30 September. The first-article check after the spindle change and the PM-2 on 9 September 2026 were both within specification.

D7. Preventive actions

- The CMMS blocks the release of new work orders on any machine that is more than 14 days beyond its PM-2 interval, in line with the escalation rule in PM-MAN-001, section 7.
- The weekly report to the Plant Manager lists every machine with a vibration or temperature warning that has lasted three operating days.
- The same check was added to the other CNC lathes at Monterrey and to the CNC lathes at Dayton and Brno.
- Gauge plans for all valve body parts were reviewed; AP-3301 and AP-3340 moved to every tenth piece.

D8. Closure and recognition

The team confirmed on 18 September 2026 that the corrective actions are in place and effective. The customer accepted the report on the same day. The team thanks the Maintenance and Quality staff of the Monterrey Plant who worked through the weekend of 15 August to contain the problem.

Approval	Name	Date
Team leader	Rafael Ocampo	18 September 2026
Plant Manager	Elena Marsh	18 September 2026
Customer	Helen Park, Brightline Hydraulics	18 September 2026